Nour Challouf

Embedded Software Engineer

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PROFILE

Electronics and Communication Systems engineer (ENET'Com, class of 2026) with solid hands-on experience in embedded C/C++: STM32 firmware, embedded Linux on Raspberry Pi, and CAN/LIN communication. I also bring a strong background in test automation, validation and CI, as well as applied machine learning for fault detection. I am looking for an embedded development or integration and validation position in Tunis or Sfax.

SKILLS

Programming languages: C, C++, Python, Java, VHDL, Matlab

Embedded systems: STM32 (CubeIDE, CubeMX), Embedded Linux on Raspberry Pi, CAN/LIN networking, DBC files, Altium Designer

Testing, validation & CI: Robot Framework, Jenkins CI/CD, Git

Web & mobile: NestJS, MongoDB Atlas, Flutter, Firebase, HTML, CSS

Machine learning & data: XGBoost, Random Forest, PyTorch, scikit-learn, Isolation Forest

Languages: Arabic (native), French (fluent), English (fluent), German (basic)

PROFESSIONAL EXPERIENCE

End-of-Studies Intern – Telnet *Feb – June 2026*

Intelligent Fault Detection & Predictive Maintenance System for UAV BLDC Motors

- Designed and built a custom multi-sensor test bench around an STM32F746ZGTx, programmed in C, with load cells, temperature, RPM and sound sensors to emulate motor faults and acquire data in real time.
- Developed an end-to-end AI pipeline covering an 8-class fault taxonomy, combining supervised classification (XGBoost) with unsupervised anomaly detection (Isolation Forest).
- Benchmarked XGBoost, Random Forest, 1D-CNN and Informer on both public and custom datasets to identify the most reliable fault detection approach.

Tools: *STM32CubeIDE, C, Python, XGBoost, Random Forest, PyTorch, scikit-learn, Git*

Summer Intern – KPIT *Aug 2025*

Automotive Lighting System: Embedded Software Integration

- Developed embedded control software in C on STM32 for a pre-built hardware prototype, implementing vehicle lighting functions driven by button inputs.
- Implemented ECU-to-ECU communication over CAN and LIN buses to ensure reliable data exchange between modules.
- Created DBC files from scratch to define the structure and meaning of the CAN messages.

Tools: *C, STM32, CAN, LIN, DBC files*

Summer Intern – Telnet *July 2025*

Automated Low-Level Test Delivery for Payment Terminals via Jenkins

- Replaced a manual SDK file-transfer process with a Python GUI application for Ingenico payment terminals.
- Integrated the solution into the existing Jenkins CI/CD pipeline.
- Validated it with Robot Framework tests run against real payment terminal functions.

Tools: *Python, Jenkins, Robot Framework, Ingenico SDK*

Summer Intern – Djagora *July – Aug 2024*

Djagora Project Management Platform: Back-End

- Built a secure back-end with NestJS, including RESTful APIs for authentication, projects and tasks.
- Designed the database schema and implemented data-access logic with MongoDB Atlas.
- Tested the APIs with Postman and used Git for version control within a team.

Tools: *NestJS, MongoDB Atlas, Postman, Git*

ACADEMIC PROJECT

Smart School Bus System – Final Year Project *2025*

Embedded Linux & Computer Vision

- Developed a dual-webcam module on Raspberry Pi (embedded Linux) running YOLOv8 for obstacle detection and OpenCV for facial recognition.
- Built a Flutter mobile application offering child check-in and real-time GPS tracking for parents.

Tools: *Raspberry Pi, Embedded Linux, YOLOv8, OpenCV, Flutter, Python*

EDUCATION

2023 – 2026 Engineering degree in Electronics & Communication Systems – ENET'Com, National School of Electronics and Telecommunications of Sfax

2021 – 2023 Preparatory cycle in Physics & Chemistry – FSS

2021 High School Diploma with Honors – Med Ali Sfax

CERTIFICATIONS & EXTRACURRICULAR ACTIVITIES

- NVIDIA Deep Learning Institute certificate
- Microsoft Certified: Azure AI Fundamentals
- **2024:** Competitor, IEEE R8 Robot Championship, Malta
- **2024 – 2025:** Webmaster, IEEE RAS ENET'Com Student Branch Chapter